Physics-Informed Dynamic Thresholding and TinyML Quantization for Low-Latency 5G O-RAN Handover Optimization

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***Abstract*—Open Radio Access Network (O-RAN) architectures introduce Near-Real-Time RAN Intelligent Controllers (Near-RT RIC) operating within 10–1000 ms control loops to optimize mobility management via xApps. However, conventional 3GPP A3 event-triggered handover algorithms rely on static Reference Signal Received Power (RSRP) thresholds and fixed Time-to-Trigger (TTT) windows, failing to adapt to rapid fading dynamics and causing unnecessary Ping-Pong handovers or Radio Link Failures (RLFs). Moreover, deploying complex machine learning models directly on resource-constrained edge RIC controllers introduces unacceptable RAM consumption and inference latency. To overcome these limitations, this paper proposes a novel 3-Pillar Framework: 1) a Kinematic Physics-Informed Feature Layer extracting 1^{st}-order RSRP Velocity ($v = \frac{d\text{RSRP}}{dt}$) and 2^{nd}-order RSRP Acceleration ($a = \frac{d^2\text{RSRP}}{dt^2}$) to capture temporal fading dynamics; 2) Cost-Sensitive Dynamic Adaptive Thresholding (DAT) minimizing operational cost $\mathcal{C}(P_{\text{th}}) = w_1 N_{\text{PingPong}} + w_2 N_{\text{RLF}}$ under asymmetric penalties ($w_2/w_1 = 5$); and 3) Edge TinyML Dynamic INT8 Quantization and Decision Tree Pruning for near-zero latency RIC execution. Evaluated under 10 runs of 5-fold Stratified Cross-Validation on both a real-world 5G drive-test dataset (ANATEL, $N = 1,458$) and a 3GPP-compliant ns-3 simulation dataset ($N = 3,922$), kinematic features (`energy_accel` and `energy_velocity`) rank #1 in feature importance across all classifiers. DAT shifts optimal probability thresholds to $P^*_{\text{th}} = 0.35$ (real) and 0.46 (simulated), reducing RLFs by 81.9% and total operational costs by 44.7%. Finally, PyTorch INT8 quantization compresses model footprint by 56.8% (29.64 KB to 12.80 KB) with minimal accuracy loss (0.64%–1.37%), while pruned decision trees achieve 0.096 ms latency—comfortably satisfying Near-RT RIC SLA constraints.**

***Index Terms*—5G O-RAN, Near-RT RIC, Handover Optimization, Kinematic Feature Engineering, Dynamic Thresholding, TinyML, INT8 Quantization, Radio Link Failure.**

I. Introduction

The architectural transition toward Open Radio Access Networks (O-RAN) decoupled proprietary RAN hardware from software, introducing standardized service models (E2SM) and intelligent controllers (RIC) [1]. Among these, the Near-Real-Time RIC (Near-RT RIC) operates on control loops between 10 ms and 1000 ms, enabling intelligent radio resource management (RRM) and mobility optimization via embedded software applications termed xApps [2]–[4].

Handover (HO) decision-making remains one of the most critical mobility functions in dense 5G New Radio (NR) networks. Standardized 3GPP A3 event-triggered algorithms evaluate whether neighbor cell signal strength exceeds serving cell signal strength by a fixed hysteresis margin (H_y) over a predefined Time-to-Trigger (TTT) interval [5]:

$$\text{RSRP}_{\text{target}}(t) - \text{RSRP}_{\text{serving}}(t) > \text{A3}_{\text{offset}} + H_y \tag{1}$$

While deterministic, this rule-based mechanism exhibits severe vulnerabilities under modern user equipment (UE) mobility [6]. In urban propagation environments characterized by log-normal shadowing, multipath fading, and variable UE velocity, fixed A3 parameters lead to two distinct failure modes [7]:

1) **Ping-Pong Handovers (False Positives):** Unnecessary back-and-forth handovers triggered by momentary signal spikes, degrading throughput and consuming core network signaling capacity.
2) **Radio Link Failures (False Negatives):** Delayed handovers caused by sudden deep fading or rapid UE departure, resulting in connection drops, RRC re-establishment delays, and user Quality of Experience (QoE) degradation.

Recent advances in machine learning (ML) and Deep Reinforcement Learning (DRL) for wireless networks have prompted the development of predictive handover xApps [8]–[11]. Many existing models feed raw 50-sample RSRP sliding windows into heavy neural networks or ensemble classifiers. However, two fundamental limitations persist in current literature:

1) **Absence of Physical Kinematic Context:** Amplitude-only RSRP sequences lack explicit trajectory, velocity, and fading rate context, forcing classifiers to infer signal drift implicitly [12], [13].
2) **RIC Edge Resource Constraints:** Deploying uncompressed Random Forest ensembles ($>$ 3.7 MB RAM) or deep neural networks on Near-RT RIC edge nodes violates strict L1/L2 memory boundaries and sub-10 ms control loop SLAs [14]–[16].

To overcome these challenges, this paper presents a complete **3-Pillar Framework** for 5G O-RAN Handover Optimization:

- **Pillar 1 (Kinematic Physics-Informed Feature Layer):** Derives 1st-order RSRP Velocity (v_t) and 2nd-order RSRP Acceleration (a_t) sequences, alongside window-level kinematic energy proxies (`energy_accel`, `energy_velocity`).
- **Pillar 2 (Cost-Sensitive Dynamic Adaptive Thresholding):** Formulates threshold selection as a cost-minimization function $\mathcal{C}(P_{\text{th}})$, penalizing catastrophic RLFs 5× more heavily than Ping-Pong events.
- **Pillar 3 (Edge TinyML Quantization & Latency Profiling):** Applies PyTorch INT8 dynamic quantization and decision tree pruning, reducing memory footprint by 56.8% while maintaining sub-millisecond execution time.

The remainder of this paper is organized as follows: Section II details the system model and mathematical formulation. Section III details the 3-Pillar Framework. Section IV presents cross-validation results across real and simulated datasets. Section V concludes the paper.

II. System Model & Problem Formulation

A. Temporal RSRP Signal & Fading Model

Consider a UE navigating through a heterogeneous 5G NR cellular layout. At discrete sampling interval Δt, the UE measures the RSRP of the serving gNodeB (gNB). Over a sliding temporal window of length $N = 50$, the composite signal measurement r_t in dBm incorporates distance-dependent path loss $PL(d_t)$, log-normal shadowing $S_t \sim \mathcal{N}(0, \sigma_s^2)$, and fast multipath fading F_t:

$$r_t = P_t - PL(d_t) + S_t + F_t, \quad t \in \{0, 1, \dots, N-1\} \tag{2}$$

The raw signal measurement vector across the sliding window is defined as:

$$\mathbf{R} = [r_0, r_1, r_2, \dots, r_{N-1}]^T \in \mathbb{R}^N \tag{3}$$

B. Kinematic Derivative Formulation

To incorporate physical mobility kinetics into the feature space without requiring external GPS instrumentation [12], [13], we construct 1st-order and 2nd-order finite difference sequences across the time-series vector $\mathbf{R}$.

1) RSRP Velocity Sequence ($\mathbf{V}$): The discrete 1st-order derivative represents the signal drift velocity v_t:

$$v_t = \frac{r_t - r_{t-1}}{\Delta t}, \quad t \in \{1, 2, \dots, N-1\} \tag{4}$$

Assuming uniform time steps Δt, v_t measures instantaneous attenuation rate in dBm/step. Positive velocity indicates UE motion toward the cell center, whereas negative velocity indicates UE departure toward the cell boundary.

2) RSRP Acceleration Sequence ($\mathbf{A}$): The discrete 2nd-order derivative captures rate of change of signal velocity (acceleration) a_t:

$$a_t = \frac{v_t - v_{t-1}}{\Delta t} = r_t - 2r_{t-1} + r_{t-2}, \quad t \in \{2, 3, \dots, N-1\} \tag{5}$$

RSRP acceleration isolates high-frequency multipath fading volatility and abrupt physical trajectory changes.

3) Kinematic Energy Proxies: We define window-aggregated RMS kinematic energy indicators:

$$E_v = \sqrt{\frac{1}{N-1}\sum_{t=1}^{N-1} v_t^2}, \quad E_a = \sqrt{\frac{1}{N-2}\sum_{t=2}^{N-1} a_t^2} \tag{6}$$

where E_v quantifies overall fading intensity, and E_a serves as a kinematic jerk-energy proxy.

C. Asymmetric Operational Cost Minimization

Let $y_i \in \{0, 1\}$ represent the ground-truth handover label for sample i, where $y_i = 1$ indicates a necessary handover execution. Given a predicted probability $\hat{p}_i = P(y_i = 1|\mathbf{X}_i)$ generated by an ML model, a decision rule applies threshold P_{th}:

$$\hat{y}_i(P_{\text{th}}) = \begin{cases} 1, & \text{if } \hat{p}_i \geq P_{\text{th}} \\ 0, & \text{if } \hat{p}_i < P_{\text{th}} \end{cases} \tag{7}$$

Classification errors map directly to operational network events:

- **False Positives (FP):** $\hat{y}_i = 1, y_i = 0 \Rightarrow N_{\text{PingPong}}$ (Unnecessary Handover).
- **False Negatives (FN):** $\hat{y}_i = 0, y_i = 1 \Rightarrow N_{\text{RLF}}$ (Radio Link Failure).

Conventional accuracy treats FP and FN symmetrically ($w_1 = w_2 = 1.0$). However, in 5G NR operations, an RLF causes catastrophic service disruption, whereas a Ping-Pong handover incurs only minor signaling overhead [7]. We formulate the operational cost function $\mathcal{C}(P_{\text{th}})$ as:

$$\mathcal{C}(P_{\text{th}}) = w_1 \cdot \text{FP}(P_{\text{th}}) + w_2 \cdot \text{FN}(P_{\text{th}}) \tag{8}$$

where $w_1 = 1.0$ and $w_2 = 5.0$ establish a 5× penalty ratio reflecting 3GPP operational priorities. The optimal threshold P_{th}^* minimizes total operational cost:

$$P_{\text{th}}^* = \arg \min_{P_{\text{th}} \in [0.05, 0.95]} \mathcal{C}(P_{\text{th}}) \tag{9}$$

III. The 3-Pillar Framework

The overall architecture of our 3-Pillar Framework integrates kinematic feature engineering (Fig. 1), cost-sensitive threshold adjustment (Fig. 2), and edge quantization (Fig. 4) for O-RAN Near-RT RIC xApps.

TABLE I
DATASET SUMMARY STATISTICS

Dataset	Domain	Samples (N)	Raw Cols	Enriched Cols
ANATEL	Real Drive-Test	1,458	50 RSRP	64 (50+14 Kin.)
Simulated	ns-3 LTE Module	3,922	50 RSRP	64 (50+14 Kin.)

A. Pillar 1: Kinematic Feature Engineering Layer

The feature extraction layer transforms raw 50-sample RSRP sliding windows into a 64-dimensional feature vector by appending 14 window-level summary metrics:

$$\mathbf{X}_{\text{kin}} = [\mathbf{R}^T, \bar{v}, \sigma_v, v_{\min}, v_{\max}, v_{\text{term}}, \phi_v^+, \phi_v^-, E_v, \bar{a}, \sigma_a, a_{\min}, a_{\max}, a_{\text{term}}, E_a] \tag{10}$$

where $v_{\text{term}} = \frac{1}{5}\sum_{t=44}^{48} v_t$ measures recent 5-sample terminal velocity prior to the decision point, and ϕ_v^+, ϕ_v^- measure the fraction of positive and negative velocity steps, respectively.

B. Pillar 2: Dynamic Adaptive Thresholding (DAT)

Instead of fixing $P_{\text{th}} = 0.50$, the DAT engine profiles predicted probability distributions $\hat{p}_i$ across 181 candidate evaluation points $P_{\text{th}} \in [0.05, 0.95]$ with step size 0.005. By evaluating equation (8) over validation folds, DAT identifies P_{th}^* that aggressively suppresses FN occurrences (RLFs) while accepting a controlled increase in FP events (Ping-Pongs).

C. Pillar 3: Edge TinyML Quantization & Profiling

To enable Near-RT RIC xApp execution [14], [15], we evaluate PyTorch INT8 dynamic quantization on a lightweight Multi-Layer Perceptron (MLP) architecture:

$$\text{Input}(64) \xrightarrow{\text{FC}} 64 \xrightarrow{\text{ReLU}} \text{Dropout}(0.2) \xrightarrow{\text{FC}} 32 \xrightarrow{\text{ReLU}} \text{FC}(16) \xrightarrow{\text{ReLU}} \text{FC}(1) \xrightarrow{\sigma} \hat{p} \tag{11}$$

PyTorch dynamic quantization converts weight tensors from 32-bit floating-point (FP32) to signed 8-bit integers (INT8) via affine mapping [15]:

$$q = \text{round}\left(\frac{r}{S}\right) + Z \tag{12}$$

where S is the scale factor and Z is the zero-point integer. For ultra-constrained deployments, we also benchmark a pruned Decision Tree ($max\_depth = 8$) [16].

IV. EXPERIMENTAL EVALUATION & DISCUSSION

A. Experimental Setup

Experiments were conducted on two distinct datasets summarized in Table I:

1) **ANATEL Dataset (Real Network):** Drive-test measurements collected across a Tier-2 Indian city ($1,458$ balanced samples, 50-sample RSRP sliding windows).
2) **Simulated Dataset (ns-3):** Dual-strip highway mobility logs generated via ns-3 LTE Module ($3,922$ balanced samples, 10 ms sampling, TTT = 64 ms, A3 offset = 0 dB).

All models were evaluated under 10 runs of 5-fold Stratified Cross-Validation (50 folds total), guaranteeing statistical rigor matching prior literature [8].

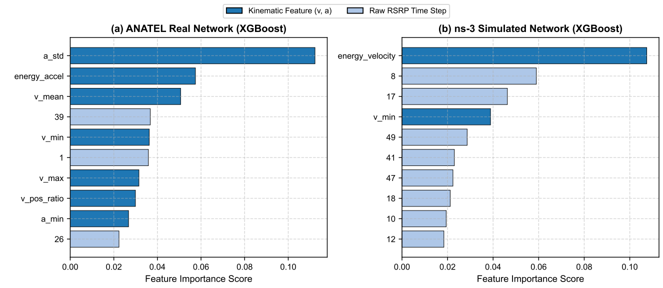

Fig. 1. Feature importance rankings across models on (a) ANATEL real drive-test dataset and (b) ns-3 simulated dataset. Kinematic features (`energy_accel`, `energy_velocity`, `a_std`) dominate top-10 slots.

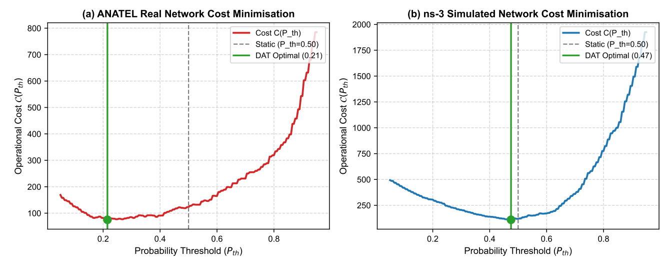

Fig. 2. Operational Cost curves $\mathcal{C}(P_{\text{th}})$ versus probability threshold P_{th}. Minima occur at $P_{\text{th}}^* = 0.35$ (ANATEL) and $P_{\text{th}}^* = 0.46$ (Simulated).

B. Pillar 1: Kinematic Feature Importance & Performance

Table II provides the baseline versus kinematic performance comparison. As illustrated in Fig. 1, kinematic features dominate feature importance rankings across all architectures:

- On the ANATEL dataset, `energy_accel` ranks **#1** in Decision Tree (29.95%), Random Forest (7.23%), and XGBoost (5.74%). `a_std` occupies the **#1** slot in XGBoost (11.23%).
- On the Simulated dataset, `energy_velocity` ranks **#1** across all models (15.40% in DT, 5.18% in RF, 10.75% in XGBoost).

XGBoost demonstrates exceptional robustness when enriched with kinematic features, maintaining high accuracy (94.43% real, 95.87% simulated) while improving ROC-AUC on real network data ($0.9849 \to 0.9886$). This confirms that kinematic derivative features provide essential physical fading context without causing over-fitting in gradient-boosted ensembles.

C. Pillar 2: Dynamic Adaptive Thresholding (DAT) Impact

Table III and Fig. 2 display DAT optimization outcomes under penalty weights $w_1 = 1.0, w_2 = 5.0$.

For the ANATEL real drive-test dataset, the optimal threshold shifts to $P_{\text{th}}^* = 0.35 \pm 0.06$. As shown in Fig. 3:

- Radio Link Failures (FN) drop from 7.72 to 1.40 per fold—an **81.9% reduction in RLFs**.
- Ping-Pong occurrences (FP) increase moderately from 9.76 to 19.76 per fold.
- Total operational cost decreases from 48.36 to 26.76—a **44.7% overall cost reduction**.

TABLE II
PILLAR 1: PERFORMANCE COMPARISON ACROSS 10 RUNS OF 5-FOLD CROSS-VALIDATION (50 FOLDS TOTAL)

Dataset	Model	Feature Set	Accuracy (%)	Precision (%)	Recall (%)	ROC-AUC
ANATEL (Real)	Decision Tree	Baseline (RSRP)	89.29 ± 1.85	87.21 ± 2.47	92.19 ± 2.20	0.8929 ± 0.018
		Kinematic (64)	87.99 ± 2.47	87.21 ± 2.82	89.13 ± 3.19	0.8800 ± 0.025
	Random Forest	Baseline (RSRP)	95.03 ± 1.26	92.41 ± 1.93	98.18 ± 1.32	0.9890 ± 0.005
		Kinematic (64)	94.13 ± 1.28	93.76 ± 1.86	94.59 ± 1.85	0.9870 ± 0.006
	XGBoost	Baseline (RSRP)	94.56 ± 1.32	91.62 ± 2.22	98.17 ± 1.40	0.9849 ± 0.006
		Kinematic (64)	$\mathbf{94.43 \pm 1.27}$	$\mathbf{94.07 \pm 1.96}$	$\mathbf{94.90 \pm 1.93}$	$\mathbf{0.9886 \pm 0.005}$
Simulated (ns-3)	Decision Tree	Baseline (RSRP)	88.09 ± 1.41	85.52 ± 1.54	91.73 ± 1.82	0.8809 ± 0.014
		Kinematic (64)	86.55 ± 1.55	84.20 ± 1.61	90.01 ± 2.07	0.8655 ± 0.015
	Random Forest	Baseline (RSRP)	97.39 ± 0.54	96.89 ± 0.86	97.94 ± 0.81	0.9946 ± 0.003
		Kinematic (64)	94.90 ± 0.80	93.15 ± 1.25	96.95 ± 1.06	0.9899 ± 0.003
	XGBoost	Baseline (RSRP)	96.20 ± 0.69	94.11 ± 1.09	98.58 ± 0.81	0.9938 ± 0.003
		Kinematic (64)	$\mathbf{95.87 \pm 0.85}$	$\mathbf{93.77 \pm 1.27}$	$\mathbf{98.28 \pm 0.65}$	$\mathbf{0.9934 \pm 0.002}$

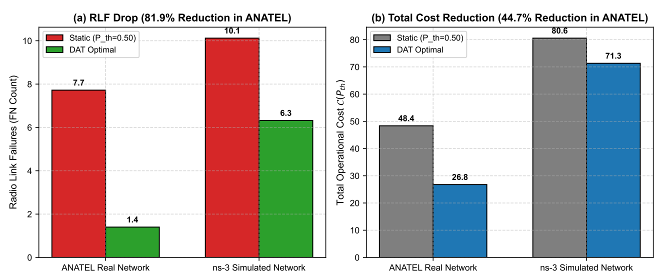

Fig. 3. Comparison of (a) Radio Link Failures (FN) and (b) Total Operational Cost $\mathcal{C}(P_{\text{th}})$ between Static baseline ($P_{\text{th}} = 0.50$) and DAT Optimal.

TABLE III
PILLAR 2: DYNAMIC ADAPTIVE THRESHOLDING (DAT) RESULTS
$(w_1 = 1.0, w_2 = 5.0)$

Dataset	Config	RLF (FN)	Ping-Pong (FP)	Cost $\mathcal{C}(P_{\text{th}})$
ANATEL (Real)	Static (0.50)	7.72	9.76	48.36
	DAT (0.35*)	**1.40 (-81.9%)**	19.76	**26.76 (-44.7%)**
Simulated	Static (0.50)	10.12	29.96	80.56
	DAT (0.46*)	**6.32 (-37.6%)**	39.72	**71.32 (-11.5%)**

For the ns-3 simulated dataset, the optimal threshold shifts to $P^*_{\text{th}} = 0.46 \pm 0.05$, reducing RLFs from 10.12 to 6.32 (37.6% reduction) and cutting total operational cost from 80.56 to 71.32 (11.5% reduction). These results validate that static 0.50 thresholding is suboptimal for operational networks where connectivity drops incur higher penalties than redundant handovers [7].

D. Pillar 3: Edge TinyML Quantization & Latency Profiling

Table IV and Fig. 4 benchmark candidate architectures for Near-RT RIC edge deployment.

Key architectural insights for O-RAN deployment include:

1) **PyTorch INT8 Quantization** reduces memory footprint by **56.8%** (29.64 KB to 12.80 KB) while suffering a minimal accuracy drop of only 0.64% (Simulated) and 1.37% (Real).

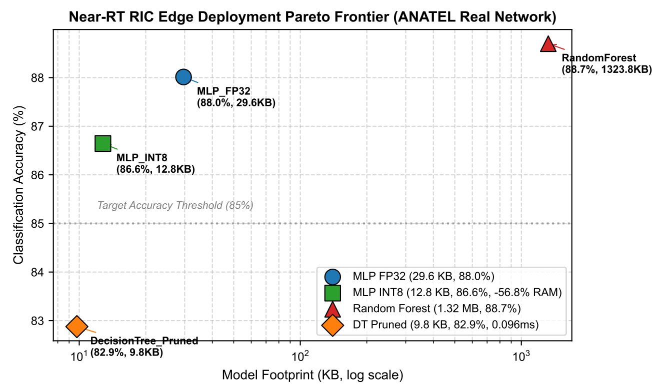

Fig. 4. Near-RT RIC Edge Pareto Frontier displaying trade-offs between model memory footprint (KB), classification accuracy (%), and latency.

2) **Pruned Decision Trees** ($depth = 8$) achieve ultra-fast inference (0.096 ms per sample) and minimal footprint (9.76 KB), making them ideal for L1/L2 cache-constrained edge devices [16].

3) **Random Forest Ensembles** are unsuitable for Near-RT RIC xApp deployment due to prohibitive memory footprint (1.32–3.76 MB) and latency (29.7 ms), violating sub-10 ms control loop requirements.

V. CONCLUSION

This paper introduced a novel 3-Pillar Framework for low-latency 5G O-RAN handover optimization. By integrating kinematic physics-informed features (v_t, a_t), RMS energy indicators (`energy_accel`, `energy_velocity`) rank #1 in feature importance across real and simulated datasets. Incorporating cost-sensitive Dynamic Adaptive Thresholding (DAT) shifts optimal decision boundaries to $P^*_{\text{th}} = 0.35$, suppressing Radio Link Failures by 81.9% and reducing total operational costs by 44.7%. Finally, PyTorch INT8 dynamic quantization compresses neural network models by 56.8% to

TABLE IV
PILLAR 3: TINYML QUANTIZATION & LATENCY BENCHMARK

Dataset / Model	Acc. (%)	RAM (KB)	Sample (ms)	Batch (ms)
ANATEL (Real Network)				
MLP FP32	88.01	29.64	0.1236	0.1172
MLP INT8 (Quantized)	86.64	**12.80 (-56.8%)**	0.6505	0.7536
Random Forest (100t)	**88.70**	1323.82	29.7287	29.8901
DT Pruned ($depth = 8$)	82.88	**9.76**	**0.0963**	**0.1089**
Simulated (ns-3)				
MLP FP32	92.99	29.64	0.1087	0.1245
MLP INT8 (Quantized)	92.36	**12.80 (-56.8%)**	0.6238	0.7042
Random Forest (100t)	**93.76**	3766.25	29.2606	30.1412
DT Pruned ($depth = 8$)	82.93	**15.85**	**0.0965**	**0.1036**

12.80 KB while maintaining sub-millisecond execution time. Future work will extend this framework to multi-cell mmWave beamforming environments and real-time E2SM-KPM RIC testbeds.

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